AA 256 Design of Composite Structures

Homework #1

Due date: Friday October 9th, 5pm



1) Show the stress transformation matrix $[T_x]$ for a rotation about the x-axis, as well as $[T_\gamma]$ for a rotation about the y-axis. Note that $[Tz]$ for a rotation about the z-axis is given in the course reader Appendix I equation (6).

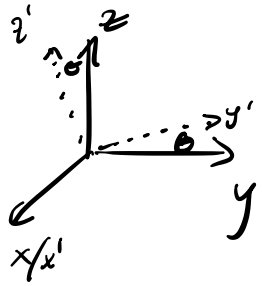
2) The matrix below gives a stress state in the X_1–X_2–X_3 coordinate system. Find the stress components after three successive rotations of the coordinate system: first 90° about X_1, then 30° about the rotated X_2 axis, and finally 45° about the rotated X_3 axis. Use the right-hand rule for each rotation. Repeat the calculation by treating the same matrix as a strain tensor. Express the transformed stresses and engineering strains in vector form, and give the corresponding stress and engineering-strain transformation matrices.

$$\sigma = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 5 \\ 5 & 5 & 6 \end{bmatrix} psi$$

$$\epsilon = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 5 \\ 5 & 5 & 6 \end{bmatrix}$$

3) Show the two-dimensional stiffness matrix based on the plane stress and plane strain assumption, respectively for Steel.

1) Show the stress transformation matrix $[T_x]$ for a rotation about the x-axis, as well as $[T_y]$ for a rotation about the y-axis. Note that $[Tz]$ for a rotation about the z-axis is given in the course reader Appendix I equation (6).



$$x' = x$$
$$y' = y\cos\theta + z\sin\theta$$
$$z' = -y\sin\theta + z\cos\theta$$

$$[R_x] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$$

$$[\sigma'] = [\beta][\sigma][\beta]^T$$

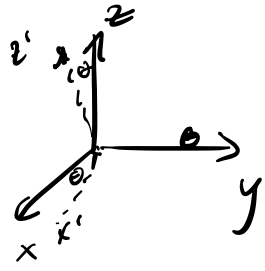
$$\begin{bmatrix} \sigma_{11}' & \sigma_{12}' & \sigma_{13}' \\ \sigma_{12}' & \sigma_{22}' & \sigma_{23}' \\ \sigma_{13}' & \sigma_{23}' & \sigma_{33}' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

↓ Symbolic Matlab

$$= \begin{bmatrix} \sigma_{11} & \sigma_{12}\cdot\cos\theta + \sigma_{13}\cdot\sin\theta & \sigma_{13}\cos\theta - \sigma_{12}\sin\theta \\ \cdot & \sigma_{22}\cdot\cos^2\theta + 2\sigma_{23}\cos\theta\sin\theta + \sigma_{33}\sin^2\theta & \cos\theta\left(\sigma_{23}\cdot\cos\theta + \sigma_{33}\sin\theta\right) - \sin\theta\left(\sigma_{12}\cos\theta + \sigma_{23}\sin\theta\right) \\ \cdot & \cdot & \sigma_{33}\cdot\cos^2\theta - 2\sigma_{23}\cos\theta\sin\theta + \sigma_{22}\sin^2\theta \end{bmatrix}$$

if $\{\sigma\} = \{\sigma_{11}\ \sigma_{22}\ \sigma_{33}\ \sigma_{23}\ \sigma_{13}\ \sigma_{12}\}^T$

$$[T_x] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta & 0 & 0 \\ 0 & \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta & 0 & 0 \\ 0 & -\cos\theta\sin\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta & 0 & 0 \\ 0 & 0 & 0 & 0 & \cos\theta & -\sin\theta \\ 0 & 0 & 0 & 0 & \sin\theta & \cos\theta \end{bmatrix}$$



$$x' = x\cos\theta - z\sin\theta$$
$$y' = y$$
$$z' = x\sin\theta + z\cos\theta$$

$$[R_y] = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$[\sigma'] = [\beta][\sigma][\beta]^T$$

$$\begin{bmatrix} \sigma_{11}' & \sigma_{12}' & \sigma_{13}' \\ \sigma_{12}' & \sigma_{22}' & \sigma_{23}' \\ \sigma_{13}' & \sigma_{23}' & \sigma_{33}' \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

↓ Symbolic Matlab

$$= \begin{bmatrix} \sigma_{11}\cos^2\theta - 2\sigma_{13}\cos\theta\sin\theta + \sigma_{33}\sin^2\theta & \sigma_{12}\cos\theta - \sigma_{23}\sin\theta & \sigma_{13}(\cos^2\theta - \sin^2\theta) + \sigma_{11}\cos\theta\sin\theta - \sigma_{33}\cos\theta\sin\theta \\ \text{''} & \sigma_{22} & \sigma_{23}\cos\theta + \sigma_{12}\sin\theta \\ \text{''} & \text{''} & \sigma_{11}\sin^2\theta + 2\sigma_{13}\cos\theta\sin\theta + \sigma_{33}\cos^2\theta \end{bmatrix}$$

if $\{\sigma\} = \{\sigma_{11}\ \sigma_{22}\ \sigma_{33}\ \sigma_{23}\ \sigma_{13}\ \sigma_{12}\}^T$

$$[T_y] = \begin{bmatrix} \cos^2\theta & 0 & \sin^2\theta & 0 & -2\cos\theta\sin\theta & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \sin^2\theta & 0 & \cos^2\theta & 0 & 2\cos\theta\sin\theta & 0 \\ 0 & 0 & 0 & \cos\theta & 0 & \sin\theta \\ \cos\theta\sin\theta & 0 & -\cos\theta\sin\theta & 0 & \cos^2\theta - \sin^2\theta & 0 \\ 0 & 0 & 0 & -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

2) The matrix below gives a stress state in the X_1–X_2–X_3 coordinate system. Find the stress components after three successive rotations of the coordinate system: first 90° about X_1, then 30° about the rotated X_2 axis, and finally 45° about the rotated X_3 axis. Use the right-hand rule for each rotation. Repeat the calculation by treating the same matrix as a strain tensor. Express the transformed stresses and engineering strains in vector form, and give the corresponding stress and engineering-strain transformation matrices.

$$\sigma = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 5 \\ 5 & 5 & 6 \end{bmatrix} psi$$

$$\epsilon = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 5 \\ 5 & 5 & 6 \end{bmatrix}$$

$$\beta_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 90^\circ & \sin 90^\circ \\ 0 & -\sin 90^\circ & \cos 90^\circ \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\beta_2 = \begin{bmatrix} \cos 30^\circ & 0 & -\sin 30^\circ \\ 0 & 1 & 0 \\ \sin 30^\circ & 0 & \cos 30^\circ \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \sqrt{3}/2 \end{bmatrix}$$

$$\beta_3 = \begin{bmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cos^2 90^\circ & \sin^2 90^\circ & 2\sin 90^\circ \cos 90^\circ & 0 & 0 \\ 0 & \sin^2 90^\circ & \cos^2 90^\circ & -2\sin 90^\circ \cos 90^\circ & 0 & 0 \\ 0 & \cos 90^\circ \sin 90^\circ & -\sin 90^\circ \cos 90^\circ & \cos^2 90^\circ - \sin^2 90^\circ & 0 & 0 \\ 0 & 0 & 0 & 0 & \cos 90^\circ & -\sin 90^\circ \\ 0 & 0 & 0 & 0 & \sin 90^\circ & \cos 90^\circ \end{bmatrix}$$

$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} \cos^2 30^\circ & 0 & \sin^2 90^\circ & 0 & -2\cos 90^\circ \sin 30^\circ & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \sin^2 30^\circ & 0 & \cos^2 90^\circ & 0 & 2\cos 30^\circ \sin 30^\circ & 0 \\ 0 & 0 & 0 & \cos 30^\circ & 0 & \sin 30^\circ \\ \cos 90^\circ \sin 90^\circ & 0 & -\cos 90^\circ \sin 30^\circ & 0 & \cos^2 90^\circ - \sin^2 30^\circ & 0 \\ 0 & 0 & 0 & -\sin 30^\circ & 0 & \cos 30^\circ \end{bmatrix}$$

$\cos 30^\circ = \frac{\sqrt{3}}{2}$

$\sin 30^\circ = \frac{1}{2}$

$$T_2 = \frac{1}{4} \begin{bmatrix} 3 & 0 & 1 & 0 & -2\sqrt{3} & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 \\ 1 & 0 & 3 & 0 & 2\sqrt{3} & 0 \\ 0 & 0 & 0 & 2\sqrt{3} & 0 & 2 \\ \sqrt{3} & 0 & -\sqrt{3} & 0 & 2 & 0 \\ 0 & 0 & 0 & -2 & 0 & 2\sqrt{3} \end{bmatrix}$$

$$T_3 = \begin{bmatrix} \cos^2 45^\circ & \sin^2 45^\circ & 0 & 0 & 0 & 2\cos 45^\circ \sin 45^\circ \\ \sin^2 45^\circ & \cos^2 45^\circ & 0 & 0 & 0 & -2\cos 45^\circ \sin 45^\circ \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos 45^\circ & -\sin 45^\circ & 0 \\ 0 & 0 & 0 & \sin 45^\circ & \cos 45^\circ & 0 \\ -\sin 45^\circ \cos 45^\circ & \sin 45^\circ \cos 45^\circ & 0 & 0 & 0 & \cos^2 45^\circ - \sin^2 45^\circ \end{bmatrix}$$

$\cos(45^\circ) = \frac{\sqrt{2}}{2}$

$$T_3 = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{2} & -\sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T = T_3 T_2 T_1$$

$$[T] = \frac{1}{8}\begin{bmatrix} 3 & 1 & 4 & 4 & 4\sqrt{3} & 2\sqrt{3} \\ 3 & 1 & 4 & -4 & -4\sqrt{3} & 2\sqrt{3} \\ 2 & 6 & 0 & 0 & 0 & -4\sqrt{3} \\ -\sqrt{6} & \sqrt{6} & 0 & -2\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ \sqrt{6} & -\sqrt{6} & 0 & -2\sqrt{6} & 2\sqrt{2} & -2\sqrt{2} \\ -3 & -1 & 4 & 0 & 0 & -2\sqrt{3} \end{bmatrix}$$

T =

0.3750	0.1250	0.5000	0.5000	0.8660	0.4330
0.3750	0.1250	0.5000	-0.5000	-0.8660	0.4330
0.2500	0.7500	0.0000	-0.0000	0.0000	-0.8660
-0.3062	0.3062	-0.0000	-0.6124	0.3536	0.3536
0.3062	-0.3062	-0.0000	-0.6124	0.3536	-0.3536
-0.3750	-0.1250	0.5000	0.0000	0.0000	-0.4330

$$\sigma = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 5 \\ 5 \\ 2 \end{bmatrix} \rightarrow \sigma' = [T]\sigma$$

$$\sigma' = \begin{bmatrix} 11.4462 \\ -2.2141 \\ 0.7679 \\ 0.0254 \\ -2.6136 \\ 1.3840 \end{bmatrix}$$

Checked w/ $\sigma' = \beta_3 \beta_2 \beta_1 \sigma \beta_1^T \beta_2^T \beta_3^T$:

```
sigma_prime (Rotation matrices):
   11.4462    1.3840   -2.6136
    1.3840   -2.2141    0.0254
   -2.6136    0.0254    0.7679
```

$e = [R]\varepsilon \qquad e' = [R]\varepsilon' = [R][T]\varepsilon = [R][T][R]^{-1}e$

$e' = [R]\varepsilon'$

$R = \mathrm{diag}([1\ 1\ 1\ 2\ 2\ 2]) \quad \therefore \quad [T]^{-T} = [R][T][R]^{-1}$

$$[T]^{-T} = \frac{1}{8}\begin{bmatrix} 3 & 1 & 4 & 2 & 2\sqrt{3} & \sqrt{3} \\ 3 & 1 & 4 & -2 & -2\sqrt{3} & \sqrt{3} \\ 2 & 6 & 0 & 0 & 0 & -2\sqrt{3} \\ -2\sqrt{6} & 2\sqrt{6} & 0 & -2\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ 2\sqrt{6} & -2\sqrt{6} & 0 & -2\sqrt{6} & 2\sqrt{2} & -2\sqrt{2} \\ -6 & -2 & 8 & 0 & 0 & -2\sqrt{3} \end{bmatrix}$$

$$\varepsilon = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 5 \\ 5 \\ 2 \end{bmatrix} \qquad e = [R]\cdot\varepsilon = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 10 \\ 10 \\ 4 \end{bmatrix} \quad \rightarrow \quad e' = [T]^{-T}e$$

$$e' = \begin{bmatrix} 11.4462 \\ -2.2141 \\ 0.7679 \\ 0.0508 \\ -5.2271 \\ 2.7679 \end{bmatrix}$$

3) Show the two-dimensional stiffness matrix based on the plane stress and plane strain assumption, respectively for Steel.

Isotropic: (steel)

Plane stress

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{13}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix}\lambda+2\mu & \lambda & \lambda & 0 & 0 & 0\\ \lambda & \lambda+2\mu & \lambda & 0 & 0 & 0\\ \lambda & \lambda & \lambda+2\mu & 0 & 0 & 0\\ 0 & 0 & 0 & \mu & 0 & 0\\ 0 & 0 & 0 & 0 & \mu & 0\\ 0 & 0 & 0 & 0 & 0 & \mu\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ \varepsilon_{23}\\ \varepsilon_{13}\\ \varepsilon_{12}\end{bmatrix} \qquad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \qquad \mu = G = \frac{E}{2(1+\nu)}$$

$$\sigma_{11} = (\lambda+2\mu)\varepsilon_{11} + \lambda\varepsilon_{22} + \lambda\varepsilon_{33}$$

$$\sigma_{22} = (\lambda+2\mu)\varepsilon_{22} + \lambda\varepsilon_{11} + \lambda\varepsilon_{33}$$

$$\sigma_{33} = (\lambda+2\mu)\varepsilon_{33} + \lambda\varepsilon_{11} + \lambda\varepsilon_{22} = 0$$

$$(\lambda+2\mu)\varepsilon_{33} = -\lambda\varepsilon_{11} - \lambda\varepsilon_{22}$$

$$\varepsilon_{33} = -\frac{\lambda}{(\lambda+2\mu)}(\varepsilon_{11}+\varepsilon_{22})$$

$$\therefore$$

$$\sigma_{11} = (\lambda+2\mu)\varepsilon_{11} + \lambda\varepsilon_{22} - \frac{\lambda^2}{(\lambda+2\mu)}(\varepsilon_{11}+\varepsilon_{22})$$

$$= \left(\lambda+2\mu-\frac{\lambda^2}{\lambda+2\mu}\right)\varepsilon_{11} + \left(\lambda-\frac{\lambda^2}{\lambda+2\mu}\right)\varepsilon_{22}$$

$$\sigma_{23} = \varepsilon_{23} = 0$$

$$\sigma_{13} = \varepsilon_{13} = 0$$

$$\sigma_{12} = \varepsilon_{12}\mu$$

$$\lambda+2\mu = \frac{E\nu}{(1+\nu)(1-2\nu)} + \frac{E}{(1+\nu)}\frac{(1-2\nu)}{(1-2\nu)}$$

$$= \frac{E}{(1+\nu)(1-2\nu)}\left[\nu + 1 - 2\nu\right]$$

$$= \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}$$

(1) $$= \frac{E(1-\nu)}{(1+\nu)(1-2\nu)} - \frac{(1+\nu)(1-2\nu)}{(1-\nu)E}\cdot\frac{E\nu}{(1+\nu)(1-2\nu)}\cdot\frac{E\nu}{(1+\nu)(1-2\nu)}$$

$$= \frac{E}{(1+\nu)(1-2\nu)}\left(1-\nu-\frac{\nu^2}{(1-\nu)}\right) = \frac{E}{(1+\nu)(1-2\nu)}\left(\frac{1-2\nu}{1-\nu}\right)$$

$$= \frac{E}{(1+\nu)(1-\nu)} = \frac{E}{1-\nu^2}$$

$$\nu = \frac{E\nu}{(1+\nu)(1-2\nu)} - \frac{(1+\nu)(1-2\nu)}{(1-\nu)E}\,\frac{E\nu}{(1+\nu)(1-2\nu)}\,\frac{E\nu}{(1+\nu)(1-2\nu)}$$

$$= \frac{E\nu}{(1+\nu)(1-2\nu)}\left(1 - \frac{\nu}{1-\nu}\right) = \frac{E\nu}{(1+\nu)(1-2\nu)}\left(\frac{1-\nu-\nu}{1-\nu}\right) = \frac{E\nu}{(1+\nu)(1-\nu)} = \frac{E\nu}{1-\nu^2}$$

$$\sigma_{11} = \frac{E}{1-\nu^2}\left(\varepsilon_{11} + \nu\varepsilon_{22}\right) \quad \therefore \quad \sigma_{22} = \frac{E}{1-\nu^2}\left(\nu\varepsilon_{11} + \varepsilon_{22}\right)$$

$$\sigma_{12} = \mu = \frac{E}{2(1+\nu)} = \frac{E}{(1-\nu^2)}\,\frac{(1+\nu)(1-\nu)}{E}\,\frac{E}{2(1+\nu)} = \frac{E}{1-\nu^2}\,\frac{1-\nu}{2}$$

$$[C] = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

Plane Strain

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} \lambda+2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda+2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda+2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{12} \end{bmatrix}$$

$$\sigma_{23} = \varepsilon_{23} = 0$$
$$\sigma_{13} = \varepsilon_{13} = 0$$
$$\sigma_{12} = \varepsilon_{12} \quad \text{(same)}$$

$$\sigma_{11} = (\lambda+2\mu)\varepsilon_{11} + \lambda\varepsilon_{22} + \lambda\varepsilon_{33} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}\varepsilon_{11} + \frac{E\nu}{(1+\nu)(1-2\nu)}\varepsilon_{22}$$

$$= \frac{E}{(1+\nu)(1-2\nu)}\left((1-\nu)\varepsilon_{11} + \nu\varepsilon_{22}\right)$$

$$\sigma_{22} = (\lambda+2\mu)\varepsilon_{22} + \lambda\varepsilon_{11} + \lambda\varepsilon_{33} = \frac{E}{(1+\nu)(1-2\nu)}\left((1-\nu)\varepsilon_{22} + \nu\varepsilon_{11}\right)$$

$$\sigma_{33} = (\lambda+2\mu)\varepsilon_{33} + \lambda\varepsilon_{11} + \lambda\varepsilon_{22}$$

$$\sigma_{12} = \mu = \frac{E}{2(1+\nu)} = \frac{E}{(1+\nu)(1-2\nu)}\,\frac{(1+\nu)(1-2\nu)}{E}\,\frac{E}{2(1+\nu)} = \frac{E}{(1+\nu)(1-2\nu)}\cdot\frac{1-2\nu}{2}$$

$$[C] = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix}$$

```
close all
clear
clc
```

Problem 1

```
syms s11 s22 s33 s23 s13 s12 C S

sigma = [s11 s12 s13;
         s12 s22 s23;
         s13 s23 s33];

Rx = [
    1 0 0;
    0 C S;
    0 -S C
    ];

sigma_x = Rx * sigma * Rx.';
disp("sigma_x:")
disp(simplify(sigma_x))

Ry = [
    C 0 -S;
    0 1 0;
    S 0 C
    ];
sigma_y = Ry * sigma * Ry.';
disp("sigma_y:")
disp(simplify(sigma_y))
```

```
sigma_x:
[          s11,                          C*s12 +
S*s13,                         C*s13 - S*s12]
[C*s12 + S*s13,         s22*C^2 + 2*s23*C*S + s33*S^2, C*(C*s23 + S*s33) -
S*(C*s22 + S*s23)]
[C*s13 - S*s12, C*(C*s23 - S*s22) + S*(C*s33 - S*s23),         s33*C^2 -
2*s23*C*S + s22*S^2]

sigma_y:
[        s11*C^2 - 2*s13*C*S + s33*S^2, C*s12 - S*s23, C*(C*s13 - S*s33) +
S*(C*s11 - S*s13)]
[                         C*s12 - S*s23,
s22,                             C*s23 + S*s12]
[C*(C*s13 + S*s11) - S*(C*s33 + S*s13), C*s23 + S*s12,         s33*C^2 +
2*s13*C*S + s11*S^2]
```

Problem 2

```
theta1 = deg2rad(90);
theta2 = deg2rad(30);
```

```
theta3 = deg2rad(45);

R1 = [
    1 0 0;
    0 cos(theta1) sin(theta1);
    0 -sin(theta1) cos(theta1)
    ];

R2 = [
    cos(theta2) 0 -sin(theta2);
    0 1 0;
    sin(theta2) 0 cos(theta2)
    ];

R3 = [
    cos(theta3) sin(theta3) 0;
    -sin(theta3) cos(theta3) 0;
    0 0 1
    ];

% Normal matrices
sigma = [
    1 2 5;
    2 3 5;
    5 5 6];

eps = [1 2 5; 2 3 5; 5 5 6];

sigma_prime_R = R3*R2*R1*sigma*R1'*R2'*R3';

% Transformation

c1 = cos(theta1);
c2 = cos(theta2);
c3 = cos(theta3);
s1 = sin(theta1);
s2 = sin(theta2);
s3 = sin(theta3);

Tx = [
    1 0 0 0 0 0;
    0 c1*c1 s1*s1 2*c1*s1 0 0;
    0 s1*s1 c1*c1 -2*c1*s1 0 0;
    0 c1*s1 -c1*s1 c1*c1-s1*s1 0 0;
    0 0 0 0 c1 -s1;
    0 0 0 0 s1 c1
    ];

Ty = [
    c2*c2 0 s2*s2 0 -2*c2*s2 0;
    0 1 0 0 0 0;
    s2*s2 0 c2*c2 0 2*c2*s2 0;
    0 0 0 c2 0 s2;
    c2*s2 0 -c2*s2 0 c2*c2-s2*s2 0;
```

```
    0 0 0 -s2 0 c2
    ];

Tz = [
    c3*c3 s3*s3 0 0 0 2*c3*s3;
    s3*s3 c3*c3 0 0 0 -2*c3*s3;
    0 0 1 0 0 0;
    0 0 0 c3 -s3 0;
    0 0 0 s3 c3 0;
    -c3*s3 c3*s3 0 0 0 c3*c3-s3*s3
    ];

T = Tz*Ty*Tx;

disp("T:")
disp(T)

sigma_vec = [1 3 6 5 5 2]';

sigma_prime = T*sigma_vec;

disp("sigma_prime (T):")
disp(sigma_prime)

disp("sigma_prime (Rotation matrices):")
disp(sigma_prime_R)

% Epsilon


eps_vec = [1 3 6 5 5 2]';
R = diag([1,1,1,2,2,2]);
e_vec = R*eps_vec;

T_eps = R * T / R;


e_prime = T_eps*e_vec;
e_vec_check = (T') \ e_vec;

disp("e_prime (T):")
disp(e_prime)

disp("e_vec_check (T^{-T}):")
disp(e_vec_check)
```

T:

0.3750	*0.1250*	*0.5000*	*0.5000*	*0.8660*	*0.4330*
0.3750	*0.1250*	*0.5000*	*-0.5000*	*-0.8660*	*0.4330*
0.2500	*0.7500*	*0.0000*	*-0.0000*	*0.0000*	*-0.8660*
-0.3062	*0.3062*	*-0.0000*	*-0.6124*	*0.3536*	*0.3536*
0.3062	*-0.3062*	*-0.0000*	*-0.6124*	*0.3536*	*-0.3536*
-0.3750	*-0.1250*	*0.5000*	*0.0000*	*0.0000*	*-0.4330*

```
sigma_prime (T):
   11.4462
   -2.2141
    0.7679
    0.0254
   -2.6136
    1.3840

sigma_prime (Rotation matrices):
   11.4462    1.3840   -2.6136
    1.3840   -2.2141    0.0254
   -2.6136    0.0254    0.7679

e_prime (T):
   11.4462
   -2.2141
    0.7679
    0.0508
   -5.2271
    2.7679

e_vec_check (T^{-T}):
   11.4462
   -2.2141
    0.7679
    0.0508
   -5.2271
    2.7679
```

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